Type I calibration of a mixed-model interaction test in an aggregated N-of-1 simulation: mean-model misfit absorbed by REML and a cluster-robust remedy

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Contents

Abstract **2**

1 Introduction **3**

2 Diagnostic instruments **4**

2.1 A first-order decomposition of the rejection rate 4

2.2 The calibration ratio under a misspecified mean and covariance 4

2.3 A two-factor model ladder 5

2.4 The sample-size gradient 5

3 Worked example **6**

3.1 Design and data-generating process 6

3.2 The analysis model 6

3.3 Replicates and Monte Carlo error 7

4 Results **8**

4.1 The model-based test is conservative because its standard error is too large . . 8

4.2 The excess comes mainly from the mean model 9

4.3 The excess does not shrink with the number of participants 12

4.4 A cluster-robust standard error restores the size 12

4.5 The remedy at an alternative cell 14

5 Discussion **14**
5.1 What the example shows 14
5.2 Agreement with the companion test-procedure paper and Kenward-Roger . . . 15
5.3 The marginal-model alternative 16
5.4 Relation to the recommended analysis 17
5.5 Limitations 17

6 Conclusions **17**

7 Reproducibility **18**

Glossary **18**

Abbreviations **19**

8 References **19**

Abstract

Background. A linear mixed model with a random intercept and a continuous-time autoregressive residual is a common analysis of an aggregated N-of-1 trial. Its model-based Wald test of a biomarker-by-drug interaction can be miscalibrated. We diagnosed why, in one simulated design.

Methods. We simulated the null of the Hybrid design with $N = 70$ participants. We computed the population calibration ratio κ exactly from the data-generating mean and covariance. We crossed the mean model with the working covariance in a model ladder. We compared model-based, CR2 cluster-robust and Kenward-Roger inference, and generalized estimating equations.

Results. The model-based standard error exceeded the empirical standard deviation by 5.1 to 7.3 percent. The Type I error was 0.033 to 0.037 against 0.05. The analysis mean, a linear trend plus a drug indicator, could not follow the curved mean trajectories. Restricted maximum likelihood absorbed this misfit into the variance parameters. Under the project's current data-generating process the exact population κ of the production model was 1.055 to 1.061 with the fitted mean. It was 1.006 with a correct mean and 1.005 to 1.008 with a free mean per visit. The calibrated appearance of the random-intercept-only model was a cancellation of two errors. The CR2 test rejected at 0.044 to 0.051.

Conclusions. In this example the conservatism came mainly from the mean model, not from the residual correlation. A visit-level mean removed most of it. A CR2 standard error restored the size without changing the model.

1 Introduction

The linear mixed model is a standard tool for testing whether a covariate moderates an effect in longitudinal data[1,2]. The quantity of interest is a fixed-effect interaction coefficient. Inference usually proceeds by a Wald test: the estimate divided by a model-based standard error, referred to a t distribution. The model-based standard error is consistent only when the working model is correct. Two parts of the working model matter. One is the within-participant covariance, the random effects together with the residual correlation. The other is the mean model.

The consequences of a misspecified working covariance are well established. The model-based standard error is then inconsistent, and the sandwich estimator repairs it[3–8]. Gurka, Edwards and Muller[9] proved that an underspecified covariance biases mixed-model fixed-effect inference in large samples as well as small ones. More participants do not remove the bias. Lu and Mehrotra[10] and Gosho et al.[11] compared the sandwich, its small-sample corrections and the Kenward-Roger method under misspecified covariance in longitudinal trials. Luke[12] evaluated the Type I error of Kenward-Roger and Satterthwaite tests in linear mixed models.

Less attention has gone to the mean model. When the fixed effects cannot follow the true mean trajectory, the residuals carry a systematic pattern. If that pattern is common to many participants, restricted maximum likelihood (REML) treats it as covariance. The variance parameters then absorb it, and the model-based standard error of a fixed effect changes. This paper shows that this mechanism, not the residual correlation structure, produced most of a conservative interaction test in one simulation of aggregated N-of-1 trials.

The example comes from the pmsimstats-ng project. The project studies power to detect biomarker-treatment interactions in aggregated N-of-1 trials under carryover[13]. Methods for combining N-of-1 trials, including the choice of covariance, are reviewed by Zucker, Ruthazer and Schmid[14] and by Araujo, Julious and Senn[15]. The companion carryover paper[16] recommends an analysis, G9, that uses a CR2 cluster-robust standard error. The companion test-procedure paper[17] reports a conservative model-based mixed model in a different design. The present paper supplies the diagnosis behind that conservatism and behind the CR2 remedy.

We make three contributions, each confined to the example. First, we compute the population calibration ratio κ exactly from the data-generating mean and covariance. This separates the mean and covariance contributions without Monte Carlo error. Second, we show that the mean misfit, absorbed by REML, accounts for most of the excess κ, and that the apparent calibration of a random-intercept-only model is a cancellation of errors. Third, we report the CR2 remedy, Kenward-Roger and generalized estimating equations on the same simulated datasets, with paired Monte Carlo standard errors. Section 2 sets out the instruments. Section 3 describes the example. Section 4 reports the results. Section 5 discusses them.

2 Diagnostic instruments

2.1 A first-order decomposition of the rejection rate

Let $\hat{\theta}$ estimate an interaction coefficient θ, with model-based standard error $\widehat{\mathrm{SE}}$ and denominator degrees of freedom ν. The Wald test rejects $H_0 : \theta = 0$ at level α when $|T| > t_{\nu,1-\alpha/2}$, with $T = \hat{\theta}/\widehat{\mathrm{SE}}$. Write

$$\hat{\theta} = \mu + \sigma Z^\star, \qquad \mu = \mathbb{E}(\hat{\theta}), \quad \sigma = \mathrm{SD}(\hat{\theta}). \tag{1}$$

Then $Z^\star$ has mean zero and unit variance by construction. Let $\kappa = \mathbb{E}(\widehat{\mathrm{SE}})/\sigma$, the calibration ratio. If the variability of $\widehat{\mathrm{SE}}$ about its mean is neglected,

$$T \approx \frac{1}{\kappa}(Z^\star + \delta), \qquad \delta = \mu/\sigma, \tag{2}$$

and

$$\alpha_{\mathrm{real}} \approx \Pr\big(|Z^\star + \delta| > \kappa\, t_{\nu,1-\alpha/2}\big). \tag{3}$$

Four quantities appear: the standardized bias δ, the scale κ, the degrees of freedom ν and the law of $Z^\star$. When $\hat{\theta}$ is approximately normal, $\delta = 0$ and ν is large, the expression reduces to the closed form

$$\alpha_{\mathrm{real}} \approx 2\,\Phi\big(-\kappa\, t_{\nu,1-\alpha/2}\big). \tag{4}$$

This is a first-order heuristic, not an exact or exhaustive decomposition. Under the approximation, $\mathrm{SD}(T) \approx 1/\kappa$ because $\mathrm{SD}(Z^\star) = 1$. Any gap between $\mathrm{SD}(T)$ and $1/\kappa$ comes from the variability of $\widehat{\mathrm{SE}}$ and from its dependence on $\hat{\theta}$, which the approximation discards. We therefore report a fifth diagnostic, the coefficient of variation of the standard error, $\mathrm{CV}(\widehat{\mathrm{SE}})$. When it is large, the closed form can fail even with $\kappa = 1$. Section 4 shows two such cases. The components are standard simulation performance measures: bias, the ratio of model-based to empirical standard error, and the rejection rate[18]. The bias δ acts as the noncentrality of a noncentral t[19]. Approximate degrees of freedom ν follow Satterthwaite[20], Welch[21] and Kenward and Roger[22].

Two quantities are measured by simulation in addition to the rejection rate. The degrees-of-freedom channel is checked by recomputing the p-values against a normal reference. The shape of the null distribution is checked by the empirical distribution function of the p-values, which should be uniform.

2.2 The calibration ratio under a misspecified mean and covariance

Let $\hat{\theta} = \ell^\top \hat{\beta}$ for a contrast vector ℓ, with $\hat{\beta}$ the generalized least squares estimate under a working covariance V_0 and fixed-effects design X. Clusters are independent, so every cross-product below is a sum over participants. The model-based variance is

$$v_0 = \ell^\top \big(X^\top V_0^{-1} X\big)^{-1} \ell. \tag{5}$$

The true sampling variance is the sandwich

$$v = \ell^\top (X^\top V_0^{-1} X)^{-1} (X^\top V_0^{-1} S V_0^{-1} X)(X^\top V_0^{-1} X)^{-1} \ell \tag{6}$$

[5,7,23]. Here S is the second moment of the outcome about the fitted working mean. When the mean model is correct, $S = V$, the true covariance. When it is not,

$$S = V + rr^\top, \tag{7}$$

where r is the misfit, the true mean minus the best-fitting working mean. In a design where every participant on a randomization path shares the same visit schedule, r is common to the participants of the path. REML estimates the variance parameters of V_0 by fitting S, not V. A common misfit pattern therefore enters the estimated variance components. The calibration ratio is $\kappa^2 = v_0/v$, with V_0 at the pseudo-true values that minimize the Kullback-Leibler criterion $\log|V_0| + \mathrm{tr}(V_0^{-1}S)$[6,24,25].

We compute κ exactly in this way from the data-generating process (DGP). At the null the biomarker is independent of the outcome with mean zero, so the information for the biomarker terms separates from the rest. The computation needs only the per-path mean and covariance of the outcome, which the DGP supplies in closed form. It has no Monte Carlo error. It is a per-participant functional, so it does not depend on the number of participants. It captures the asymptotic calibration and excludes finite-sample REML terms by construction. We validated it against the simulation in Section 4.

2.3 A two-factor model ladder

The ladder refits the same datasets under a mean model crossed with a working covariance. The covariance factor has four rungs: ordinary least squares (M0), a random intercept (M1), a continuous-time first-order autoregressive residual alone (M2), and both (M3, the production model). The mean factor has four levels: the fitted mean of the analysis model, the fitted mean plus expectancy terms (the mean structure of the project's F1 analysis), a free mean for each visit, and a correct mean. The correct mean is a free mean for each path-by-visit cell in simulation, and the true mean in the analytic computation. Because every rung is fitted to the same data, the rungs can be compared replicate by replicate.

2.4 The sample-size gradient

The analytic κ of Section 2.2 does not depend on the number of participants. A simulation gradient in N therefore cannot show whether the excess is asymptotic. That is settled by the analytic computation. The gradient checks only for finite-sample terms in the REML variance estimates, the terms that the Kenward-Roger correction addresses[22,26]. We report the slope of the simulated κ on $\log_2 N$ with a confidence interval.

3 Worked example

3.1 Design and data-generating process

Table 1 summarizes the study in the ADEMP structure of Morris, White and Crowther[18]. The design is the Hybrid design: open-label titration, blinded discontinuation, then a brief crossover. It has four randomization paths and eight visits, at weeks 4, 8, 9, 10, 11, 12, 16 and 20. $N = 70$ is the total across paths, allocated 18, 18, 17 and 17. The outcome Y_{it} is a baseline level minus three components: a natural-history component, a placebo-response component whose standard deviation scales with the design expectancy η_t, and a drug-response component. Each component follows a modified Gompertz mean curve and a first-order autoregressive correlation in calendar weeks with $\rho = 0.7$. The components are non-negative reductions in symptom severity. An increase in any component therefore lowers Y_{it}, and interaction coefficients are negative under the alternative.

Carryover acts on the drug-response mean after discontinuation, with a half-life of $t_{1/2} = 1$ week. The interaction is carried by the correlation between the biomarker and the drug response, c_{bm} (the covariance-moderation architecture of the companion DGP paper[27]). At $c_{bm} = 0$ the interaction is null, and the mean-moderation and covariance-moderation architectures generate identical data under a common seed (verified, `check-null-architecture-equivalence.R`).

Two versions of the DGP appear in this paper. The **pre-correction run** of 2026-05-20 used a cross-component correlation of $c_1 = 0.2$ at the same visit but $c_\times = 0.1$ at different visits, a mismatch documented in the project's driver audit. It also used the original carryover rule, which compounds the decay. The **extended DGP** of the companion DGP paper sets $c_1 = c_\times = 0.2$, lets carryover decay from the last on-drug mean, and stops on a covariance matrix that is not positive definite. Every table caption names the version that produced it. The analytic computation is reported for both. No cell of either version needed a positive-definiteness repair. The minimum eigenvalue of the covariance matrix over the four paths is 0.104 for the pre-correction constants and 0.184 for the extended DGP at $c_{bm} = 0$.

3.2 The analysis model

The production analysis model is

$$Y_{it} = \beta_0 + \beta_B B_i + \beta_T w_t + \beta_D D^*_{it} + \theta\, B_i D^*_{it} + u_i + \varepsilon_{it}, \tag{8}$$

with B_i the biomarker, drawn from $\mathcal{N}(0,1)$, w_t the calendar week, $u_i \sim \mathcal{N}(0, \sigma^2_u)$ a participant random intercept, and ε_{it} a residual with continuous-time first-order autoregressive correlation $\rho_a^{|w_t - w_s|}$ and variance σ^2_ε. It is fitted by REML with `nlme::lme`[28,29], with `corCAR1(form = ~t | ptID)`. The drug term D^*_{it} defines three specifications. Under E1 it is the binary drug state D_{it}. Under E2 it is the exposure-decayed indicator $D_{bc,it}$ at the true half-life. E3 uses D_{it} and adds the lagged just-off-drug indicator L_{it}. The interaction θ is the coefficient of `bm:Db`

Table 1: ADEMP summary of the simulation study.

Element	Content
Aims	Explain the conservatism of the model-based interaction test and evaluate remedies.
Data-generating mechanisms	Hybrid design, N = 70, c_bm = 0 (null) and 0.45 (alternative), half-life 1 week, rho = 0.7. Extended DGP and pre-correction run.
Estimand	theta, the biomarker-by-drug interaction coefficient. theta = 0 at the null.
Methods	E1, E2, E3 under M0 to M3 and four mean models. CR2 Satterthwaite, G9, F1. GEE with Liang-Zeger and Mancl-DeRouen sandwiches. Kenward-Roger on M2.
Performance measures	Rejection rate with MCSE and binomial test, kappa, CV of the SE, mean estimate with MCSE, paired differences between procedures.

under E1 and E3 and of `bm:Dbc` under E2. This is the historical analysis of the project, not its current recommendation.

In the codes of the companion carryover paper[16], E1 with model-based inference is G1 and with CR2 inference is G6. E2 at the true half-life is G3, and G8 with CR2. G9 is the recommended analysis: E2 with the analysis half-life chosen by AIC over $\{0.25, 0.5, 1, 2\}$ weeks, a REML refit and a CR2 test. F1 adds the design expectancy and its interaction with the biomarker, `bm + t + Dbc + bm:Dbc + De + bm:De`, at a fixed analysis half-life of 0.2 week, with a CR2 test.

3.3 Replicates and Monte Carlo error

The decomposition and the ladder used 5,000 null replicates, the CR2 and alternative-cell runs 3,000, generalized estimating equations 5,000, the Kenward-Roger check 2,000 and the gradient 1,500 per point. All null-cell runs at $N = 70$ share one seed, so replicate i is the same dataset in every run. Every comparison between procedures is therefore paired. The gradient runs draw their own datasets, including at $N = 70$. A rejection rate $\hat{\pi}$ from n replicates has Monte Carlo standard error (MCSE) $\sqrt{\hat{\pi}(1-\hat{\pi})/n}$. A paired difference of rejection indicators has MCSE equal to the standard deviation of the differences over $\sqrt{n}$. The MCSE of κ is approximated by $\kappa/\sqrt{2(n-1)}$, which ignores the variability of the mean standard error.

Table 2: First-order decomposition at the primary null cell (Hybrid, $N = 70$, $c_{bm} = 0$), production model M3 with the fitted mean. Extended DGP ($c_\times = 0.2$, carryover decaying from the last on-drug mean). Fits that failed are excluded, not refitted without the residual correlation. Closed form is the closed-form size of Section 2.1 at the observed κ and df.

Spec	n	Mean est. (MCSE)	κ	CV(SE)	SD(T)	Type I (MCSE)	Binom. p	Type I, z	Closed form	df
E1	5000	+0.029 (0.012)	1.054	0.092	0.945	0.0370 (0.0027)	< 0.001	0.0370	0.038	487
E2	5000	+0.034 (0.016)	1.073	0.093	0.929	0.0330 (0.0025)	< 0.001	0.0338	0.035	487
E3	5000	+0.029 (0.012)	1.051	0.093	0.948	0.0374 (0.0027)	< 0.001	0.0376	0.039	486

4 Results

4.1 The model-based test is conservative because its standard error is too large

We fitted M3 to the primary null cell for E1, E2 and E3. Table 2 reports the decomposition. The model-based standard error exceeded the empirical standard deviation of the estimate, with $\kappa = 1.054$, 1.073 and 1.051 for E1, E2 and E3. The MCSE of each κ is about 0.011. The Type I error was 0.037, 0.033 and 0.037, each with MCSE about 0.0027. Each was below 0.05 by an exact binomial test (each $p < 0.001$).

The other first-order quantities did not produce the conservatism. The degrees of freedom were about 487, and a normal reference moved the Type I error by at most 0.0008. The closed form (4) at the observed κ predicted 0.035 to 0.039. The observed range was 0.033 to 0.037. Each prediction exceeded its observed value, by 0.5 to 0.8 MCSE. The three errors share a direction because the three specifications were fitted to the same datasets. The linearization $\alpha_{\text{real}} \approx 0.05 - 0.23(\kappa - 1)$, from the derivative $-2z_{0.975}\varphi(z_{0.975}) = -0.229$ at $\kappa = 1$, is used only to state the elasticity $\mathrm{d}\log\alpha_{\text{real}}/\mathrm{d}\log\kappa = -4.58$. At $\kappa = 1.073$ it gives 0.033, against the closed form 0.035.

The mean estimate was +0.029 to +0.034, 2.1 to 2.4 MCSE from zero. This is not a property of the estimator. At $c_{bm} = 0$ the biomarker is independent of the outcome and symmetric about its mean. The model contains the biomarker main effect and the drug main effect, so replacing B_i by its reflection about the mean changes the sign of $\hat{\theta}$ and leaves the variance estimates unchanged. Hence $\mathbb{E}(\hat{\theta}) = 0$ exactly. The displacement is one chance fluctuation. It recurs across specifications because they share datasets: the E1 and E2 estimates have correlation 0.91.

The random-intercept standard deviation fell below 1 in 2.3 percent of E1 fits, at the boundary of the parameter space, against a pseudo-true value of 9.49. Failed fits numbered 0 of the 30,000, counting three specifications and two mean models.

Figure 1 shows the null p-values. They lie below the diagonal over the whole unit interval, as an inflated standard error predicts.

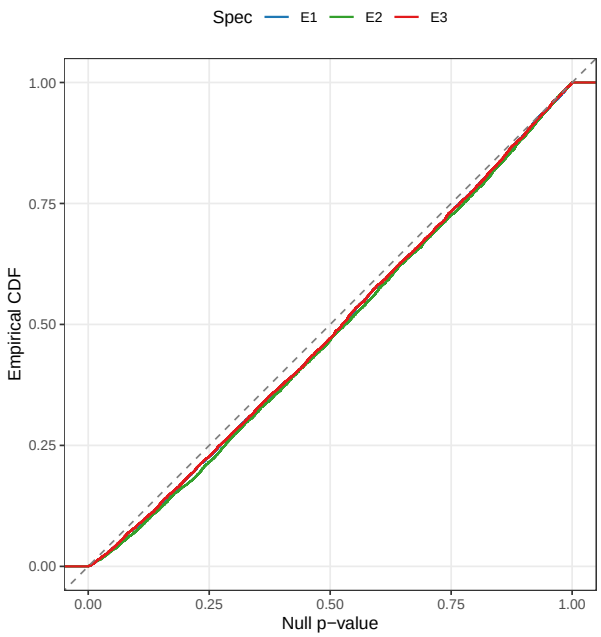


Figure 1: Empirical distribution function of the null p-values of the model-based M3 test at the primary cell, against the Uniform(0,1) diagonal. The curves lie below the diagonal across the unit interval.

4.2 The excess comes mainly from the mean model

We computed the population κ for every combination of mean model and working covariance (Table 3). The computation reproduced the simulation. For the extended DGP and E1 under M3 it gave $\kappa = 1.057$ against 1.054 simulated, $\rho_a = 0.729$ against a simulated median of 0.731, and $\sigma_u = 9.49$ against a median of 9.32. As a check of the computation, a DGP whose outcome covariance is exactly a random intercept plus an autoregressive residual returned $\kappa = 1.000$, $\sigma_u = 10.00$ and $\rho_a = 0.70$, the true values. That DGP holds the placebo-response variance constant over visits.

Table 3: Population calibration ratio κ computed exactly from the DGP mean and covariance, by mean model and working covariance (M0 independence, M1 random intercept, M2 autoregressive residual, M3 both). Pre-correction: $c_\times = 0.1$, original carryover rule. Extended: $c_\times = 0.2$, carryover decaying from the last on-drug mean. E3 is within 0.01 of E1 throughout and is in the data file.

DGP	Spec	Mean	M0	M1	M2	M3
Pre-correction	E1	Fitted	1.308	0.972	1.063	1.071
Pre-correction	E1	Fitted + expectancy	1.426	1.112	1.047	1.066
Pre-correction	E1	Free per visit	1.266	0.899	1.015	1.015
Pre-correction	E1	Correct	1.269	0.901	1.016	1.017
Pre-correction	E2	Fitted	1.269	0.949	1.132	1.082
Pre-correction	E2	Fitted + expectancy	1.354	1.059	1.111	1.081

Pre-correction	E2	Free per visit	1.236	0.888	1.093	1.024
Pre-correction	E2	Correct	1.235	0.887	1.090	1.022
Extended	E1	Fitted	1.307	0.959	1.048	1.057
Extended	E1	Fitted + expectancy	1.442	1.116	1.038	1.053
Extended	E1	Free per visit	1.266	0.886	0.999	1.005
Extended	E1	Correct	1.268	0.888	1.000	1.006
Extended	E2	Fitted	1.265	0.935	1.083	1.061
Extended	E2	Fitted + expectancy	1.362	1.058	1.070	1.058
Extended	E2	Free per visit	1.233	0.873	1.041	1.008
Extended	E2	Correct	1.232	0.872	1.038	1.006

The mean misfit was large. The standard deviation of the true mean of the outcome over the eight visits of a path was 3.94 to 4.27 points. The fitted mean, a linear trend plus a drug term, left a root-mean-square misfit of 4.18 to 4.41 points under M3. The misfit is common to the participants of a path. REML absorbed it into the variance parameters: under M3 the pseudo-true residual standard deviation was 12.36 with the fitted mean and 11.35 with the correct mean.

Under the extended DGP, M3 had $\kappa = 1.055$ to 1.061 with the fitted mean and 1.006 with the correct mean, over E1 to E3. The covariance misspecification of M3 therefore contributes about one percent at most, and the mean misfit the rest. Under the pre-correction DGP the values were 1.064 to 1.082 and 1.017 to 1.022. With a correct mean only the covariance differs between the two versions, so the larger covariance share there came from the mismatched cross-component constant. The remaining 0.006 under the extended DGP comes from the placebo-response variance, which changes with the design expectancy and so departs from a random intercept plus an autoregressive residual. With that variance held constant the computation returns $\kappa = 1$, as reported above. A free mean for each visit, common to all paths, removed nearly all of the excess: M3 gave 1.005 to 1.008. Adding the expectancy terms to the fitted mean did not: M3 gave 1.050 to 1.058. The expectancy covariate takes two values, so it cannot follow the curved natural-history and drug-response means.

A paired simulation confirmed the analytic result. In 1,487 replicates of the null cell, with $c_\times = 0.1$ and carryover decaying from the last on-drug mean, M3 for E1 was fitted twice to each dataset: once to Y_{it} and once to Y_{it} minus its true path-by-visit mean. κ fell from 1.072 to 1.014, a paired difference of 0.057 with bootstrap standard error 0.0027. The Type I error rose from 0.033 to 0.047, a paired difference of +0.014 with MCSE 0.0031. This check was run by the referee of an earlier draft (`diagnose-null-meanmisfit-referee.R`). Under the extended DGP the same check on the 5,000 replicates of Table 2 gave $\kappa = 1.054$ with the fitted mean and 1.003 with the true mean removed, a paired difference of 0.051 (bootstrap SE 0.0010). The Type I error changed by +0.010 (MCSE 0.0014).

The ladder explains the random-intercept-only model (Table 4). With the fitted mean, M1 was close to calibrated for E1 and slightly liberal for E2. Its population κ was 0.935 to 0.959, and in simulation M1 rejected at 0.054 for E1 and 0.060 (MCSE 0.0034) for E2. With a correct

Table 4: Simulated ladder at the primary null cell. Extended DGP ($c_\times = 0.2$, carryover decaying from the last on-drug mean). The same datasets are refitted under each row. M2 is fitted with `gls` under the nlminb optimizer and tight tolerances.

Mean	Model	Covariance	E1: κ / Type I (MCSE)	E2: κ / Type I (MCSE)
Fitted	M0	OLS	1.323 / 0.0090 (0.0013)	1.286 / 0.0112 (0.0015)
Fitted	M1	Random intercept	0.978 / 0.0536 (0.0032)	0.958 / 0.0602 (0.0034)
Fitted	M2	Autoregressive residual	1.043 / 0.0398 (0.0028)	1.094 / 0.0310 (0.0025)
Fitted	M3	Both (production)	1.054 / 0.0370 (0.0027)	1.073 / 0.0330 (0.0025)
Fitted + expectancy	M1	Random intercept	1.117 / 0.0274 (0.0023)	1.073 / 0.0342 (0.0026)
Fitted + expectancy	M3	Both (production)	1.047 / 0.0370 (0.0027)	1.068 / 0.0334 (0.0025)
Free per path and visit	M1	Random intercept	0.905 / 0.0724 (0.0037)	0.894 / 0.0814 (0.0039)
Free per path and visit	M3	Both (production)	1.004 / 0.0476 (0.0030)	1.019 / 0.0440 (0.0029)
Free per visit	M1	Random intercept	0.906 / 0.0750 (0.0037)	0.896 / 0.0792 (0.0038)
Free per visit	M3	Both (production)	1.004 / 0.0468 (0.0030)	1.022 / 0.0432 (0.0029)

mean its population κ was 0.872 to 0.888. M1 would then be anti-conservative, because it ignores the serial correlation. The absorbed mean misfit raises its κ by a similar amount, and the two errors cancel. The paired difference in Type I error between M1 and M3 for E1 was +0.017 (MCSE 0.0031). Removing the residual correlation from M3 does not repair the test. It trades one error for another. The simulated mean-model rungs agree with the population values. With a free mean per visit, M3 had simulated $\kappa = 1.004$ and 1.022 for E1 and E2 and rejected at 0.047 and 0.043. With the expectancy terms added to the fitted mean it had 1.047 and 1.068 and rejected at 0.037 and 0.033, no better than the fitted mean.

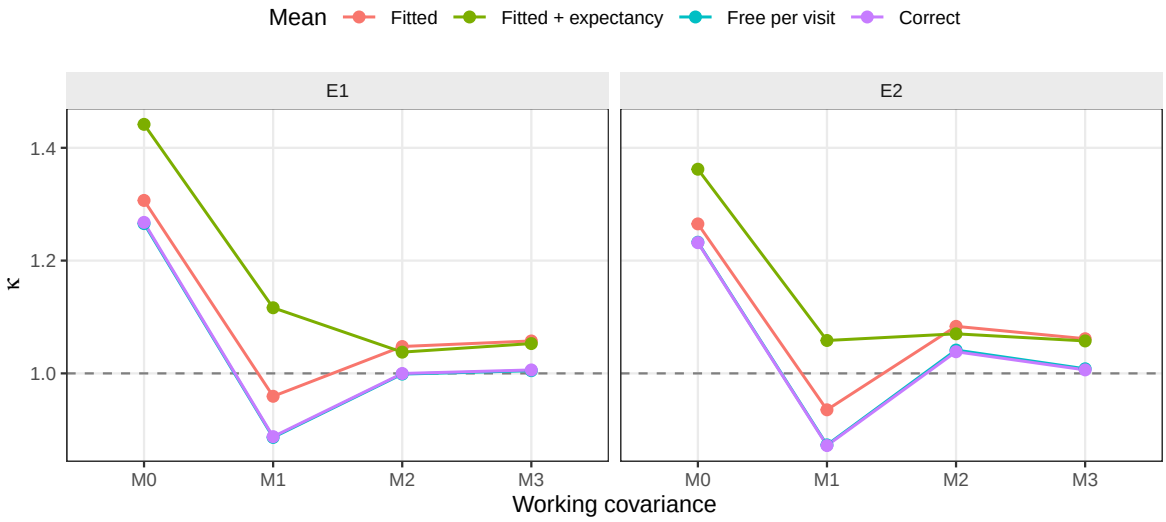


Figure 2: Population calibration ratio by working covariance and mean model, extended DGP, computed exactly. The dashed line marks calibration. With the fitted mean, M3 is conservative and M1 appears calibrated. With a correct or a per-visit mean, M3 is calibrated and M1 is anti-conservative.

M0 had κ near 1.32. Ordinary least squares charges a within-participant contrast with the between-participant variance as well[30–32]. The contrast is not purely within participant, be-

Table 5: Sample-size gradient, production model M3 with the fitted mean, 1,500 replicates per point. Extended DGP ($c_\times = 0.2$, carryover decaying from the last on-drug mean). The MCSE of each κ is about 0.02.

N	E1: κ / Type I (MCSE)	E2: κ / Type I (MCSE)	E3: κ / Type I (MCSE)
35	1.057 / 0.036 (0.005)	1.083 / 0.032 (0.005)	1.054 / 0.037 (0.005)
70	1.042 / 0.046 (0.005)	1.052 / 0.039 (0.005)	1.039 / 0.047 (0.005)
140	1.061 / 0.041 (0.005)	1.072 / 0.037 (0.005)	1.059 / 0.041 (0.005)
280	1.036 / 0.036 (0.005)	1.050 / 0.029 (0.004)	1.033 / 0.036 (0.005)

cause the on-drug fraction differs by path, five of eight visits on some paths and four of eight on others.

4.3 The excess does not shrink with the number of participants

Table 5 reports M3 at $N = 35$, 70, 140 and 280. The weighted slope of κ on $\log_2 N$ was -0.004 (95 percent interval -0.021 to 0.012) for E1, -0.008 (-0.025 to 0.009) for E2 and -0.004 (-0.021 to 0.013) for E3. The intervals include zero. They are also wide enough to include a change of 0.01 per doubling. The gradient therefore shows only that no large finite-sample REML term is present. The gradient is too coarse to resolve the size. At $N = 70$ its E1 and E3 sizes were 0.046 and 0.047, on datasets independent of Table 2, against 0.037 and 0.037 there. The differences are 1.7 gradient MCSE, consistent with chance. The evidence that the excess is asymptotic is the analytic computation of Section 4.2, which does not depend on N.

4.4 A cluster-robust standard error restores the size

We replaced the model-based standard error of M3 by the CR2 bias-reduced cluster-robust standard error, clustered on participant, with Satterthwaite degrees of freedom[20,33,34], computed with `clubSandwich`[35]. Let $\tilde{e}_i$ and $\tilde{X}_i$ be the residuals and design of participant i after the generalized-least-squares transformation $\hat{V}_0^{-1/2}$, and H_{ii} the corresponding block of the hat matrix. A cluster-robust variance is

$$\hat{V}_{\mathrm{CR}} = (\tilde{X}^\top \tilde{X})^{-1} \Big(\textstyle\sum_i \tilde{X}_i^\top A_i \, \tilde{e}_i \tilde{e}_i^\top A_i \tilde{X}_i \Big) (\tilde{X}^\top \tilde{X})^{-1}. \tag{9}$$

CR2 sets $A_i = (I - H_{ii})^{-1/2}$, which makes $\hat{V}_{\mathrm{CR}}$ unbiased when the working model is correct[34,36]. The jackknife choice $A_i = (I - H_{ii})^{-1}$ is the estimator of Mancl and DeRouen[37] in the marginal setting. The Satterthwaite degrees of freedom are $\nu = (\sum_g \ell_g)^2 / \sum_g \ell_g^2$ in the eigenvalues ℓ_g of the matrix that defines the quadratic form[34]. The point estimate and the model are unchanged.

Table 6 reports the result. The CR2 κ was 0.981 to 1.004. The CR2 Type I error was 0.0500, 0.0437 and 0.0507, at $z = +0.00$, -1.59 and +0.17 from 0.05. The paired increase over the model-based test was +0.012, +0.011 and +0.012, with paired MCSE 0.0026, 0.0028 and 0.0026. The CR2 computation failed in 0, 0 and 0 of the 3,000 replicates. These are excluded.

Table 6: CR2 remedy at the primary null cell, M3 with the fitted mean. Extended DGP ($c_\times = 0.2$, carryover decaying from the last on-drug mean). Failed counts replicates in which the fit or the CR2 computation failed. They are excluded.

Spec	n	Failed	κ model	κ CR2	CV(SE) CR2	Type I model (MCSE)	Type I CR2 (MCSE)	Binom. p CR2	CR2 − model (paired MCSE)	CR2 df (5th to 95th)
E1	3000	0	1.051	0.981	0.168	0.0380 (0.0035)	0.0500 (0.0040)	1.000	+0.0120 (0.0026)	23.7 (17.4 to 30.0)
E2	3000	0	1.076	1.004	0.169	0.0323 (0.0032)	0.0437 (0.0037)	0.121	+0.0113 (0.0028)	23.7 (17.2 to 29.9)
E3	3000	0	1.048	0.981	0.168	0.0383 (0.0035)	0.0507 (0.0040)	0.867	+0.0123 (0.0026)	23.7 (17.4 to 30.0)

Table 7: G9 and F1 at the primary null cell, CR2 inference. Extended DGP ($c_\times = 0.2$, carryover decaying from the last on-drug mean).

Procedure	n	Failed	κ CR2	Type I CR2 (MCSE)	Binom. p
F1	3000	0	0.982	0.0490 (0.0039)	0.834
G9	2986	14	0.952	0.0543 (0.0041)	0.294

The closed form (4) does not hold for CR2. At the observed CR2 κ and mean degrees of freedom it predicts 0.038 to 0.043. The coefficient of variation of the CR2 standard error was 0.17, against 0.09 for the model-based one. The Satterthwaite reference absorbs that variability with 24 degrees of freedom, 5th to 95th percentile 17 to 30, far below the model-based 487. The degrees of freedom reflect the uneven leverage of participants on the interaction contrast. The paths differ in their on-drug fraction, which is the likely source[38,39].

Table 7 adds the two analyses the project recommends. F1 rejected at 0.049 (MCSE 0.0039). G9 rejected at 0.054 (MCSE 0.0041), with 14 of 3,000 fits failed and excluded. Both are within Monte Carlo error of 0.05 (binomial $p = 0.834$ and 0.294). G9 differs from the CR2 test of E2 only by the AIC choice of the half-life. On the 2,986 replicates where both succeeded, the selection raised the size by +0.010 (paired MCSE 0.0023). The CR2 κ of G9 was 0.952: its standard error does not carry the variability that the selection adds.

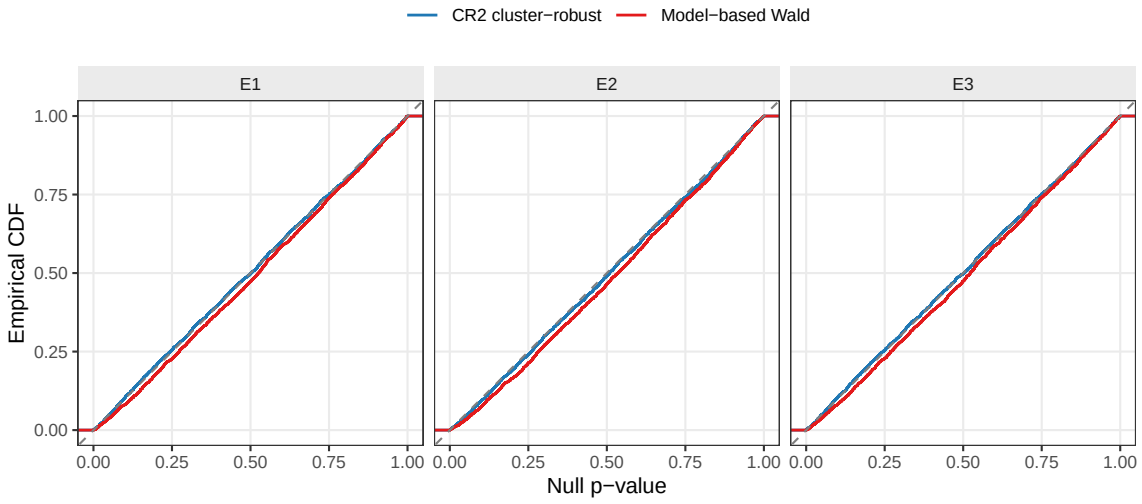


Figure 3: Null p-value distribution functions under the model-based and CR2 tests of M3, by specification. The model-based curves lie below the diagonal. The CR2 curves lie close to it.

Table 8: Alternative cell: covariance-moderation architecture, Hybrid, $N = 70$, $c_{bm} = 0.45$, half-life 1 week. Extended DGP ($c_{\times} = 0.2$, carryover decaying from the last on-drug mean). Power is the rejection rate of M3 with the fitted mean.

Spec	n	κ model	κ CR2	Power model (MCSE)	Power CR2 (MCSE)	CR2 − model (paired MCSE)
E1	3000	1.119	0.993	0.790 (0.0074)	0.812 (0.0071)	+0.021 (0.0046)
E2	3000	1.133	0.991	0.872 (0.0061)	0.894 (0.0056)	+0.021 (0.0043)
E3	3000	1.116	0.994	0.788 (0.0075)	0.809 (0.0072)	+0.021 (0.0046)

4.5 The remedy at an alternative cell

We repeated the comparison at $c_{bm} = 0.45$ under the covariance-moderation architecture, with the same design and $N = 70$ (Table 8). The model-based κ was 1.116 to 1.133 and the CR2 κ 0.991 to 0.994. CR2 raised power by +0.021, +0.021 and +0.021 for E1, E2 and E3 (paired MCSE 0.0046, 0.0043 and 0.0046).

On the 3,000 replicates in which all three specifications succeeded, the paired power advantage of E2 over E1 was +0.082 (MCSE 0.0060) under the model-based test and +0.082 (MCSE 0.0057) under CR2. The difference of these differences was +0.000 (MCSE 0.0056). The advantage of E2 over E3 was +0.084 (MCSE 0.0060) and +0.084 (MCSE 0.0058). At this cell the recalibration raised power by a similar amount for every specification and left their order unchanged. This is one cell.

5 Discussion

5.1 What the example shows

In this example the model-based interaction test of the production model was conservative. The main cause was the mean model. The fitted mean, a linear trend plus a drug term, could not follow the curved mean trajectories of the three components. REML treated the common misfit as covariance, and the variance parameters grew. The model-based standard error of the interaction grew with them. Under the extended DGP the misspecified residual covariance contributed about one percent of κ. This revises the reading of an earlier draft, which attributed the conservatism to the autoregressive residual layer.

The revision changes the practical lesson. Removing the residual correlation does not repair the test. It produces a model whose two errors offset. A mean model that follows the visits does repair it. In this design a free mean per visit brought the population κ of M3 to 1.005 to 1.008. Expectancy terms alone did not. The CR2 standard error repairs the size without changing the model. It does not repair the cause. The two remedies can be combined. These statements are confined to one DGP, one design and one estimand. That a mean misfit absorbed by REML can make a mixed-model interaction test conservative in other designs is a hypothesis, supported by the $V + rr^{\top}$ argument of Section 2.2 but not tested here.

The direction of the error depends on the model. M3 with the fitted mean was conservative. M1 with a correct mean was anti-conservative. The Liang-Zeger sandwich of Section 5.3 was

anti-conservative. Gurka, Edwards and Muller[9] showed that an underspecified covariance can inflate the Type I error, and that the bias does not shrink with the sample size. Our result is compatible with theirs. Here the conservative direction arises because the absorbed misfit inflates the variance parameters.

5.2 Agreement with the companion test-procedure paper and Kenward-Roger

The companion test-procedure paper[17] reports a model-based mixed model of the same class with κ of 1.10 to 1.16 and a size of 0.02 to 0.03, in a different design and DGP. It classifies the conservatism as a calibration defect of the working covariance, not of the degrees of freedom. It notes that Kenward-Roger or Satterthwaite adjustments cannot repair a conservative test, because they inflate the variance or lower the degrees of freedom. It recommends CR2 inference. The present results agree on each point. They refine the cause. The estimated working covariance is inflated, and in this example the inflation comes from the mean misfit that REML absorbs. We did not decompose the conservatism of that paper, whose mean model includes expectancy terms but not a free mean per visit.

Our empirical Kenward-Roger check is of the M2 rung, not of M3. Kenward-Roger is available in R for marginal models through the `mmrm` package[40], which does not fit a random intercept. SAS PROC MIXED supports Kenward-Roger with a random intercept and a spatial power residual. We did not run it. On the continuous-time exponential covariance, Kenward-Roger changed the interaction standard error by 0.04 percent on average (Table 9). On the unstructured covariance it changed it by 7.08 percent. With Kenward-Roger the unstructured covariance still rejected at 0.0610 and 0.0630 for E1 and E3, 2.26 and 2.67 MCSE above 0.05. The residual anti-conservatism agrees with Gosho et al.[11], who found Kenward-Roger with an unstructured covariance close to nominal only when the number of completers is large relative to the number of visits.

Table 9: Kenward-Roger against Satterthwaite in `mmrm` at the primary null cell. Extended DGP ($c_\times = 0.2$, carryover decaying from the last on-drug mean). The exponential fit has no random intercept: it is the M2 rung, not the production model M3. The Satterthwaite fit keeps the model-based covariance.

Covariance	Method	Spec	κ	df	Type I (MCSE)
Exponential (M2)	Kenward-Roger	E1	1.041	451	0.0430 (0.0045)
Exponential (M2)	Kenward-Roger	E2	1.096	501	0.0310 (0.0039)
Exponential (M2)	Kenward-Roger	E3	1.037	449	0.0445 (0.0046)
Exponential (M2)	Satterthwaite	E1	1.041	451	0.0430 (0.0045)
Exponential (M2)	Satterthwaite	E2	1.095	501	0.0310 (0.0039)
Exponential (M2)	Satterthwaite	E3	1.037	449	0.0445 (0.0046)
Unstructured	Kenward-Roger	E1	0.952	178	0.0610 (0.0054)
Unstructured	Kenward-Roger	E2	0.967	152	0.0520 (0.0050)
Unstructured	Kenward-Roger	E3	0.952	177	0.0630 (0.0054)

Table 10: Generalized estimating equations at the primary null cell, AR(1) working correlation in visit order. Extended DGP ($c_\times = 0.2$, carryover decaying from the last on-drug mean). LZ is the Liang-Zeger sandwich and MD the Mancl-DeRouen sandwich. Entries in the four rightmost rate columns are Type I error (MCSE). Relative efficiency is the variance of the M3 estimate over that of the GEE estimate on the same datasets.

Spec	n	κ LZ	κ MD	LZ, z	LZ, t	MD, z	MD, t	Rel. eff.
E1	5000	0.955	1.015	0.0672 (0.0035)	0.0634 (0.0034)	0.0538 (0.0032)	0.0516 (0.0031)	0.71
E2	5000	0.963	1.023	0.0682 (0.0036)	0.0634 (0.0034)	0.0544 (0.0032)	0.0508 (0.0031)	0.71
E3	5000	0.955	1.015	0.0678 (0.0036)	0.0636 (0.0035)	0.0540 (0.0032)	0.0510 (0.0031)	0.71
Unstructured		Satterthwaite		E1	0.888	178	0.0820 (0.0061)	
Unstructured		Satterthwaite		E2	0.906	152	0.0705 (0.0057)	
Unstructured		Satterthwaite		E3	0.887	177	0.0805 (0.0061)	

In the pre-correction run the `gls` fit of the M2 rung and the `mmrm` exponential fit of nominally the same model disagreed, with a ratio of mean standard errors of 1.004 for E1 and 0.981 for E2. The `gls` fit had used the `optim` optimizer with default tolerances. With the `nlminb` optimizer and tight tolerances, a check on 21 replicates of the extended DGP gave standard errors that matched `mmrm` within a relative difference of 7.5e-06 in every one of 63 fits. The earlier M2 rung had not reached the REML optimum. The corrected drivers use the tight settings. In the full extended-DGP run the ratio of mean standard errors was 1.000 over E1 to E3.

5.3 The marginal-model alternative

We fitted the generalized estimating equations counterpart of the analysis model with `geepack`[41], Gaussian family and an AR(1) working correlation (Table 10). Two features limit the comparison. First, `geepack` indexes the AR(1) lag by visit order and ignores the irregular spacing of the Hybrid visits. We checked that the reported standard error equals an inline sandwich with order lags. A calendar-time working correlation is not available in that package and was not evaluated. Second, in the pre-correction run the Mancl-DeRouen standard error came from a separate refit with `geesmv`[42]. The corrected driver computes both standard errors from one fit. In every case we report both standard errors against both references, a normal one and t on $K - p$ degrees of freedom, with K participants and p fixed effects[43].

The Liang-Zeger sandwich gave κ of 0.955 to 0.963 and rejected at 0.067 to 0.068 with a normal reference. The closed form (4) predicts 0.059 to 0.061 at those κ, below the observed rates. This is the second case where the first-order decomposition fails. The sandwich standard error is variable and correlated with the estimate[44–46]. The Mancl-DeRouen correction with a t reference rejected at 0.0516, 0.0508 and 0.0510, 0.3 to 0.5 MCSE from 0.05 (smallest binomial $p = 0.604$). At this cluster count it is calibrated within Monte Carlo error. This reverses the pre-correction run, in which it rejected at 0.055 to 0.058. That run took the

Mancl-DeRouen standard error from a separate geesmv refit and used the earlier DGP. We did not separate the two changes.

The GEE estimate was less efficient than the M3 estimate on the same data. Its empirical standard deviation for E1 was 1.014 against 0.854, a relative efficiency of 0.71. About 29 percent of the information is lost. The marginal route is therefore not equivalent to the conditional route with CR2 in this design. Both the CR2 test of M3 and the Mancl-DeRouen test with a t reference held the nominal size. Only the CR2 test keeps the efficiency of the mixed-model estimate.

5.4 Relation to the recommended analysis

The CR2 tests evaluated here are G6 (E1) and G8 (E2) in the codes of the companion carryover paper[16]. That paper recommends G9, which adds an AIC choice of the analysis half-life and reports a measurable size increment from the selection step. Here the increment was +0.010 (paired MCSE 0.0023), and G9 stayed within Monte Carlo error of 0.05. The present paper explains why the CR2 step is needed: the model-based standard error of the historical analysis is inflated by the mean misfit. The mean-model results suggest a further option, a free mean per visit, which G9 does not use. Its power cost was not evaluated here.

5.5 Limitations

The study has one DGP, one design, one estimand and one alternative cell. The analytic κ assumes equal allocation to paths, whereas the simulation allocates 18, 18, 17 and 17, and it excludes finite-sample terms by construction. The simulation tables come from the extended DGP.

The parametric-bootstrap reference and the design-dependence analyses of the examination plan (stages 4b and 5) were not run. The examination plan expected M3 to be the well-specified member of the ladder. The results show otherwise. Kenward-Roger was evaluated on M2 only. The power cost of a free mean per visit, and its behavior under the alternative, were not evaluated. The ranking result at the alternative rests on one cell. It should not be extended to crossover designs or to the mean-moderation architecture without evidence.

6 Conclusions

First, in this example the model-based Wald test of the interaction in the production mixed model was conservative. Its standard error was 5.1 to 7.3 percent too large and its size was 0.033 to 0.037.

Second, the main cause was the mean model. REML absorbed the misfit of a linear trend plus a drug term into the variance parameters. With a correct mean the population κ of M3 was 1.006.

Third, the apparent calibration of the random-intercept-only model was a cancellation of two errors. It is not evidence that the residual correlation should be dropped.

Fourth, the excess is asymptotic. The analytic κ does not depend on the number of participants. Kenward-Roger changed the standard error of the autoregressive M2 fit by about 0.04 percent. A degrees-of-freedom correction cannot repair a conservative test.

Fifth, a CR2 cluster-robust standard error with Satterthwaite degrees of freedom restored the size to within Monte Carlo error of 0.05 and raised power at the alternative cell by a similar amount for each specification. A free mean per visit removed most of the cause.

7 Reproducibility

The drivers are in `analysis/scripts/carryover-sensitivity/`. `paper10-common.R` holds the DGP settings of the corrected drivers: $c_1 = c_\times = 0.2$, carryover decaying from the last on-drug mean, and a stop on a covariance matrix that is not positive definite. Each run records the git commit and date in its output. The drivers are `diagnose-kappa-analytic.R` (the population κ), `diagnose-null-type1.R` (decomposition and correct-mean check), `diagnose-null-type1-stages46.R` (ladder and gradient), `diagnose-null-type1-cr2.R` (CR2, G9 and F1), `diagnose-alt-cell-cr2.R` (alternative cell), `diagnose-null-type1-gee.R` (generalized estimating equations), `diagnose-null-type1-kr.R` (Kenward-Roger) and `check-null-architecture-equivalence.R`. The script `run-paper10-refined.sh` runs all of them with single-threaded linear algebra and writes to `output/paper10-refined/`. The pre-correction outputs of 2026-05-20 remain in `output/` for comparison. The software versions were R 4.6.1, `nlme` 3.1-169, `clubSandwich` 0.7.0, `geepack` 1.3.13 and `mmrm` 0.3.17, cited in Sections 3 to 5. The examination plan that preceded the manuscript is retained as `type1-examination-plan.md`.

Glossary

κ. The calibration ratio: the mean model-based standard error divided by the empirical standard deviation of the estimates. Above 1 the test is conservative.

Population (pseudo-true) κ. κ computed exactly from the DGP mean and covariance, with the working variance parameters at the values REML converges to as the number of participants grows.

Mean misfit. The true mean of the outcome at a path and visit minus the best-fitting mean of the analysis model.

M0 to M3. Working covariances: independence (ordinary least squares), random intercept, continuous-time autoregressive residual alone, and both. M3 is the production model.

E1, E2, E3. Drug terms of the analysis model. E1 is the binary drug state. E2 is the exposure-decayed indicator at the true half-life. E3 is the binary drug state plus a lagged just-off-drug indicator.

G1, G3, G6, G8, G9, F1. Analysis codes of the companion carryover paper. G1 and G3 are E1 and E2 with model-based inference. G6 and G8 are the same with CR2. G9 adds an AIC choice of the analysis half-life. F1 adds the design expectancy and its interaction with the biomarker at a fixed half-life of 0.2 week.

Hybrid design. Open-label titration, blinded discontinuation, then a brief crossover, with four randomization paths.

corCAR1. The continuous-time first-order autoregressive residual correlation of `nlme`, $\rho_a^{|w_t - w_s|}$.

c_{bm}**.** The correlation between the biomarker and the drug response on drug, which carries the interaction in the covariance-moderation architecture.

c_1**,** $c_\times$**.** Cross-component correlations at the same visit and at different visits. The extended DGP sets both to 0.2. The pre-correction run used $c_\times = 0.1$. Code names `c.cf1t` and `c.cfct`.

Extended DGP. The recommended data-generating process of the companion DGP paper: separable AR(1) with $\rho = 0.7$ and $c_1 = 0.2$, a biomarker-response correlation that decays after discontinuation, carryover that decays from the last on-drug mean, and a stop on matrices that are not positive definite. Code label `refined`.

Original carryover rule. The carryover rule of the original DGP, which applies the decay to the already-carried mean of the previous visit and so compounds it.

Pre-correction run. The simulation outputs of 2026-05-20, made with $c_\times = 0.1$ and the original carryover rule.

CV(SE). The coefficient of variation of a standard error across replicates.

Paired MCSE. The Monte Carlo standard error of a difference between two procedures applied to the same datasets.

Abbreviations

AIC, Akaike information criterion. CR2, bias-reduced cluster-robust variance estimator of Bell and McCaffrey. DGP, data-generating process. GEE, generalized estimating equations. KR, Kenward-Roger. LZ, Liang-Zeger. MCSE, Monte Carlo standard error. MD, Mancl-DeRouen. MMRM, mixed model for repeated measures. OLS, ordinary least squares. REML, restricted maximum likelihood.

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